Evaluating RETFound: A Foundational Model for Glaucoma Detection using Macular OCT Images

Joseph Tran, BS[1,2]; Rohith Ravindranath, MS[2]; Robert T. Chang, MD[2]; Sophia Y. Wang, MD[2]

[1]*University of Nevada, Reno School of Medicine, Reno, NV*
[2]*Byers Eye Institute, Stanford University, Palo Alto, CA*

University of Nevada, Reno
School of Medicine

Introduction

- Glaucoma is a progressive optic neuropathy that is the leading cause of irreversible blindness worldwide. Early detection and intervention is critical to slowing disease progression and preserving vision.
- Traditional AI models require large annotated datasets and extensive computational resources to achieve high performance, making training time-consuming and resource-intensive.
- RETFound is an open-source foundation model trained on 1.6 million unlabeled retinal images, designed to be adapted for various downstream ophthalmological tasks[1].

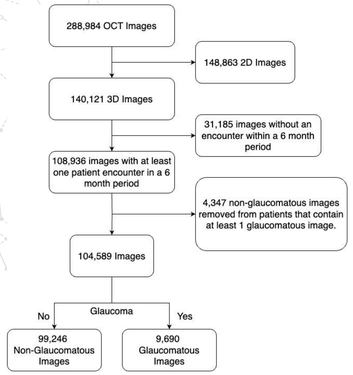
Purpose

- To evaluate RETFound with an independent clinical dataset on a new task, detecting glaucoma using macular optical coherence tomography (OCT) images, thus validating the generalizability of the model.

Methods

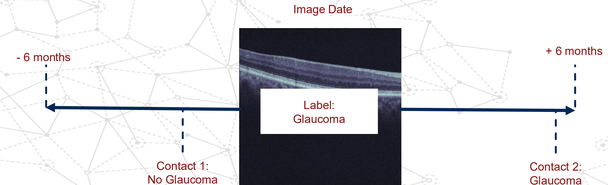
Dataset

- 104,589 OCT scans from 10,113 patient seen at Stanford Eye Clinics.
- Images are split at patient-level into 80:10:10 train, validation, and test sets.
- 2D input images was the middlemost scan from each OCT volume.



Labeling

- Glaucoma classification based on ICD-10 Codes diagnosed within 6 months of when image was taken.
- Glaucoma suspect was labeled as glaucoma negative.



Results

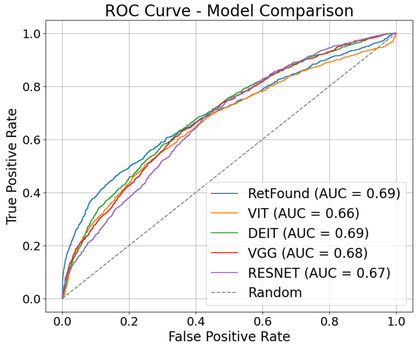
RETFound and Baseline Model Architectures

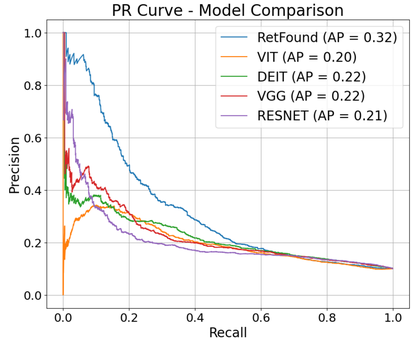
- RETFound: A masked autoencoder (MAE) foundation model pretrained on retinal image data.
- We compare RETFound to four other baseline models pretrained using ImageNet:
  - Vision Transformer (ViT)
  - Data-efficient Image Transformer (DeiT)
  - Visual Geometry Group (VGG)
  - Residual Network (RESNET)

Demographics

	Glaucoma Positive		Glaucoma Negative		Total	
	N = 826		N = 9287		N = 10113	
	Mean	Std	Mean	Std	Mean	Std
Age	74.40	18.30	65.12	19.12	65.92	19.22
	N	%	N	%	N	%
Male	434	52.54	4206	45.29	4640	45.88
Female	392	47.46	5081	54.71	5473	54.12
Non-Hispanic White	274	33.17	3574	38.48	3848	38.05
Non-Hispanic Black or African American	30	3.63	215	2.32	245	2.42
Hispanic or Latino (of any race)	149	18.04	1525	16.42	1674	16.55
Non-Hispanic AAPI	257	31.11	2497	26.89	2754	27.23
Non-Hispanic Other	98	11.86	1111	11.96	1209	11.95
Unknown / Declined to State	18	2.18	365	3.93	383	3.79

RETFound and Baseline Model Comparison

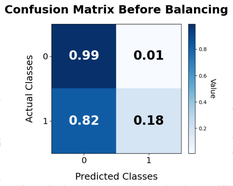


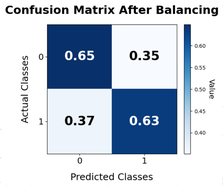


- ROC-AUC scores were comparable across models, indicating similar overall discriminative ability.
- RETFound achieved a noticeably higher precision-recall (PR) curve, suggesting superior performance in identifying positive cases.
- Models were trained using a class balanced training set.

Metrics

Model	Accuracy	AUC-ROC	F1-Score	Sensitivity	Specificity
RETFound	0.647	0.693	0.263	0.633	0.649
ViT	0.334	0.663	0.205	0.852	0.288
DeiT	0.577	0.692	0.249	0.700	0.563
VGG	0.531	0.684	0.241	0.745	0.507
RESNET	0.612	0.671	0.246	0.633	0.610

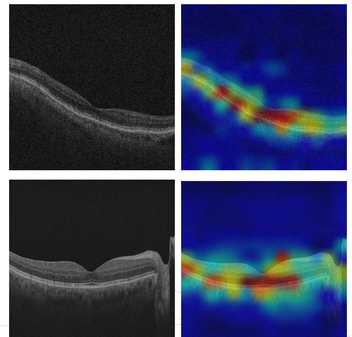




- RETFound performance improved with class balancing. Initial confusion matrix (left), post-balancing result (right).

Heatmap Visualization

- Model attention map overlaid on OCT image (right), indicating areas most influential in glaucoma prediction.
- Positive glaucoma (top), Negative glaucoma (bottom)



Conclusion and Discussion

Glaucoma Detection

- All models showed comparable AUROC (~0.7), suggesting similar overall discriminative ability for glaucoma detection.
- RETFound outperformed on the PR-Curve, suggesting it may be better at handling class-imbalanced datasets, typical in general populations.

RETFound as a Foundation Model

- Despite being pre-trained on different retinal tasks, RETFound generalized well to glaucoma detection.
- RETFound required only moderate training and minimal task-specific customization to achieve its results, suggesting feasible implementation with minimal fine-tuning.

Limitations

- Ground truth labels were derived from diagnostic codes rather than direct image-level grading.
- Images were selected based on position (middlemost of scan), without quality control for signal strength, alignment, or pathology visibility.
- Glaucoma is a progressive disease with stages ranging from early “glaucoma suspect” cases to advanced disease.

Future Work

- Incorporate clinician-reviewed annotations for improved accuracy.
- Incorporate preprocessing or quality filtering for images.
- Address multiple staging of the disease by including glaucoma suspects

References

1. Zhou Y, Chia MA, Wagner SK, et al. A foundation model for generalizable disease detection from retinal images. Nature. 2023;622(7981):156-163. doi:10.1038/s41586-023-06555-x

Email jotran@cs.stanford.edu for more information.

Authors have no conflicts of interest to disclose.